

Spectral Graph Power Dynamics: A Scoped Closure Audit

IAS2026 final paper draft | negative evidence retained

Abstract

We audit a frozen IEEE-39 SG-to-GFL benchmark using spectral and collective closure identities, transverse structure checks, and archived nonlinear evidence. The nominal P4 H4 value is reproduced at $0.1270064667 \text{ s}^{-1}$, the frozen P4 structure has 15 stable proper subsets, and four randomized algebraic test modules pass. The archived nonlinear table reports 32/32 declared TDS agreements. However, the historical V9 transfer result is 395/512 rather than the stronger expected 511/512, with 116 false-safe and 1 false-unstable cases. An isolated PowerDynamics 5.0.0 run validates only its official IEEE-39 tutorial equilibrium, not parity with the frozen GFL model. The conclusion is therefore benchmark- and policy-conditioned, with robustness and second-model generalization left untested.

1. Scope and protocol

The execution layer was created from the IAS2026_FINAL_SCIENTIFIC_EVIDENCE_FREEZE tag in a linked worktree. Historical source tables were copied read-only into raw/gate0. Expectations were treated as hypotheses: a mismatch is reported as REFUTED, and an unavailable dependent experiment is reported as NOT_TESTED or STOPPED_BY_GATE.

Primary benchmark	Frozen IEEE-39 SG-to-GFL closure model
Nominal gate	P4 H4 alpha absolute error $\leq 1\text{e-}6 \text{ s}^{-1}$
Theory tests	local/collective factorization, bridge, contextual return, transverse construction
Independent software	PowerDynamics 5.0.0 official IEEE-39 tutorial equilibrium
Labels	IEEE39_VALIDATED, THEOREM_VALIDATED, POWERDYNAMICS_VALIDATED, RETROSPECTIVE, REFUTED, NOT_TESTED, STOPPED_BY_GATE

2. Frozen IEEE-39 closure

Gate 0 reproduces the nominal and governed P4 values from the immutable final-closure tables. The nominal H4 alpha is $0.1270064666836382 \text{ s}^{-1}$; the documented governor value is $-0.07454098428813066 \text{ s}^{-1}$. The sign change is evidence of policy dependence and a local repair effect, not a universal certificate. The F7A boundary witness is $g^*=0.2076814044$ at $f=0.7064247879 \text{ Hz}$, with boundary Q sign -1 and contextual-return sign +1.

3. Algebra and nonlinear evidence

The property suite uses randomized block and transverse constructions. It tests the port identities, determinant factorization, the bridge without inversion of the disturbance matrix, the contextual Schur identity under $Q_{ii}=0$, and the transverse rotation/drift residuals. All four modules pass. The frozen G2 table reports 32 declared TDS verdict agreements and two P4 H4 unstable rows.

4. Transfer and independent software

The archived V9 reveal contains 512 portfolios: 395 correct, 116 false-safe, and 1 false-unstable. This refutes the stronger near-perfect transfer expectation and exact antichain claim. PowerDynamics reaches its official IEEE-39 tutorial equilibrium, but that tutorial uses a different model class. Same-model parity and same-mechanism reproduction are stopped by gate.

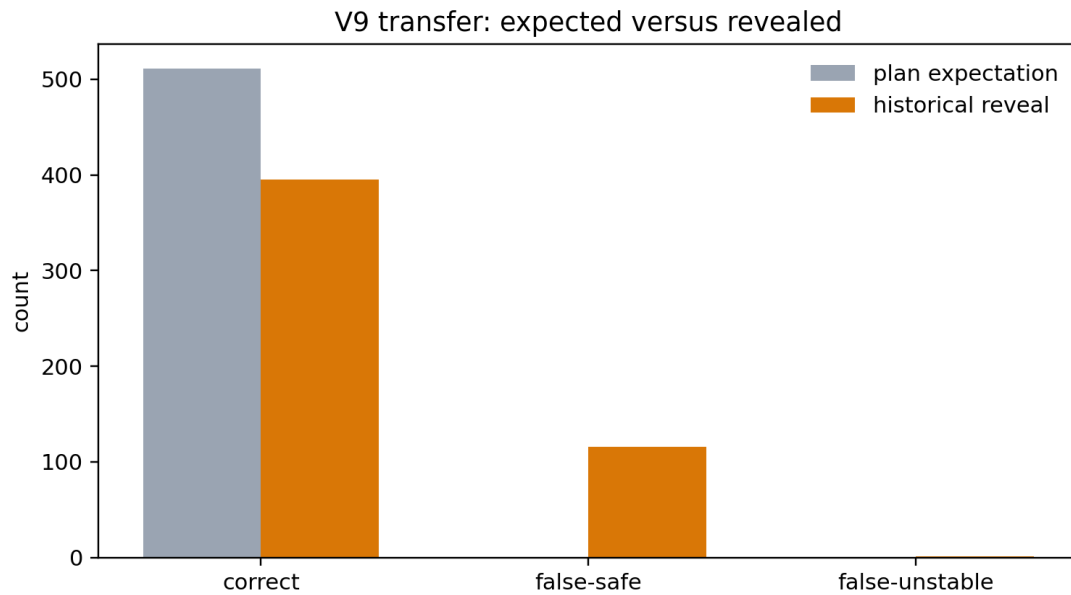


Figure 2. Historical V9 transfer expectation versus revealed result.

5. Limitations and conclusion

No physical common uncertainty envelope, robust LFT radius, second converter model, new blind topology/model holdout, new Julia TDS, EMT, current-limit, DC-link, protection, switching-path, or hardware certification is claimed. The safe conclusion is a benchmark-specific, policy-conditioned minimal incompatibility with exact frozen nominal closure and negative evidence against stronger transfer/generalization language.